EPFL Center for Digital Trust

policy brief #3

Purpose first: The need for a paradigm shift in privacy-preserving data sharing

A purpose-driven approach to privacy-enhancing technologies is needed to reconcile policy goals with technical feasibility

- → Data sharing that supports the secondary use of data while preserving individual privacy is crucial for Europe's future data economy.
- → Privacy-enhancing technologies (PETs) play a key role in addressing this challenge. However, they face a fundamental trade-off between retaining information for analysis (utility) and preventing harmful inferences (privacy) which is often not adequately recognized.
- → Current technology recommendations tend to overestimate the effectiveness of specific privacy technologies while downplaying their weaknesses.
- → A shift towards purpose-driven approaches that emphasize risk-based frameworks and realistic, empirical assessments of technological capabilities, is essential to balance privacy and utility in data sharing.

What is this about?

As data sharing becomes integral to economic, research, and public sectors, it is crucial to balance the goals of expanded data access and individual privacy protection. Privacy-enhancing technologies (PETs) have been heavily promoted as solutions to this challenge. However, contrary to the promises and assumptions made by some private vendors and policy advisors, PETs cannot fully resolve the inherent trade-off between maintaining data utility and safeguarding privacy. Current recommendations to policymakers tend to overlook these fundamental limitations of PETs. Technologies such as synthetic data generation (SDG) are often advertised as fully privacy-preserving solutions suitable for enabling widespread data sharing. But these portrayals are overly optimistic and ignore empirical evidence that questions the effectiveness of these technologies. Despite these issues, commercial and advisory bodies continue to promote a technology-centric approach to issues that are inherently political.

To align policy ambitions with technical realities, a paradigm shift is needed in our approach to privacy-preserving data sharing. Instead of relying solely on technology to resolve inherent trade-offs, it is more effective to identify the specific purpose of data sharing and to recognize the associated privacy risks of each use case. Furthermore, future technology recommendations must be reconciled with realistic expectations: While PETs can effectively minimize privacy risks and provide utility for well-defined analytical tasks, procedural controls are necessary in scenarios that involve broad exploratory data usage — as envisioned in many current data space projects — to decide whether data utility or privacy should take precedence.

Why is this important?

Adopting a purpose-driven approach to privacy-preserving data sharing is essential to achieve the goals of innovation, research, and fair data economies. While PETs play an important role in ensuring privacy, current recommendations often mistakenly assume that PETs alone can address all privacy challenges, overlooking the broader systemic requirements for effective data sharing. The effectiveness of PETs ultimately depends on the specific goals of data use and their impact on the privacy – utility trade-off. Relying solely on technology-centric data-sharing frameworks risks information leakage, inaccurate data analysis, and potential privacy breaches, as well as increasing the difficulty of effectively assessing privacy risks for practitioners.

As policy circles and the EU Commission consider developing large-scale data sharing through PETs, there is an urgent need for a realistic, purpose-driven assessment of the proposed solutions. This includes a more critical evaluation of the recommended privacy technologies, emphasizing the need for greater scientific and technical input to guide informed and effective policy decisions. Without this change, future data-sharing systems may not meet their utility goals or fail to protect people's sensitive information. Given the complex challenge of balancing data utility with individual privacy, it is essential to develop practical strategies for ensuring that data is shared safely and responsibly, rather than solely relying on assurances that technology will address all issues.

The findings in this brief are based on C4DT Insight #3 *Purpose first: The need for a paradigm shift in privacy-preserving data sharing*, 2025.

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Policy recommendations

→ **Recommendation 1**
Recognize and accept the inherent trade-off in data sharing
Policymakers and data practitioners should acknowledge the fundamental trade-off between data utility and privacy.

→ **Recommendation 2**
Define clearly the purpose of data sharing first
Data-sharing practices should be driven by a precise definition of the data-sharing purpose, before selecting the appropriate PETs to implement these goals.

→ **Recommendation 3**
Evaluate technical claims critically and realistically
PETs should not be seen as panaceas. Depending on the intended purpose of the data-sharing scenario, no PET might be able to entirely mitigate privacy risks. Data practitioners should critically evaluate the claims made by technical-solution vendors and avoid accepting blanket promises about privacy-utility balancing technologies.

→ **Recommendation 4**
Develop standards to evaluate the inherent trade-off
Instead of over-relying on future technological solutions, policymakers should prioritize the development of guidance in the form of standards that determine the prioritization of privacy or utility.

→ **Recommendation 5**
Shift to purpose-led, risk-based approach to data sharing
Policymakers should support the establishment of a purpose-led, risk-based approach to data-sharing that evaluates and manages inherent risks, based on the specific objectives of data use.

→ **Recommendation 6**
Encourage tailored approaches
Practitioners should be encouraged to evaluate privacy technologies in the context of specific data use cases by considering their purpose and utility requirements.

→ **Recommendation 7**
Enhance transparency and accountability in the data-sharing supply chain
Policymakers should enhance transparency and accountability in the data-sharing supply chain by requiring vendors to disclose technology blind spots and risk strategies, thus encouraging the sharing of privacy responsibilities between vendors and data owners while building trust.

→ **Recommendation 8**
Support ongoing dialogue and research
To ensure that policies are informed by the latest empirical research and technological developments, policymakers and data practitioners should remain engaged in continuous dialogue with researchers about the evolving capabilities and limitations of PETs.